



Advanced Digital Image Processing Lecture #8

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Contents

- ❖ Introduction to Image Enhancement
- ❖ Image Enhancement in the Frequency Domain



Introduction to Image Enhancement

- ❖ Enhancement is to process an image so that the result is more suitable than the original image for a specific application
- ❖ Two approaches use for image enhancement:
 - ❖ Spatial domain: Refer to the image plane itself. Direct manipulation of pixels in an image
 - ❖ Frequency Domain: Based on modifying the Fourier Transform of an image

Image Enhancement in the Frequency Domain

- ❖ Any continuous function $f(x)$ can be expressed as an integral over sine and/or cosine waves of different frequencies, each multiplied by a different coefficient.
- ❖ Any signal that is non-zero over a finite range can also be represented by an infinite number of sine waves.

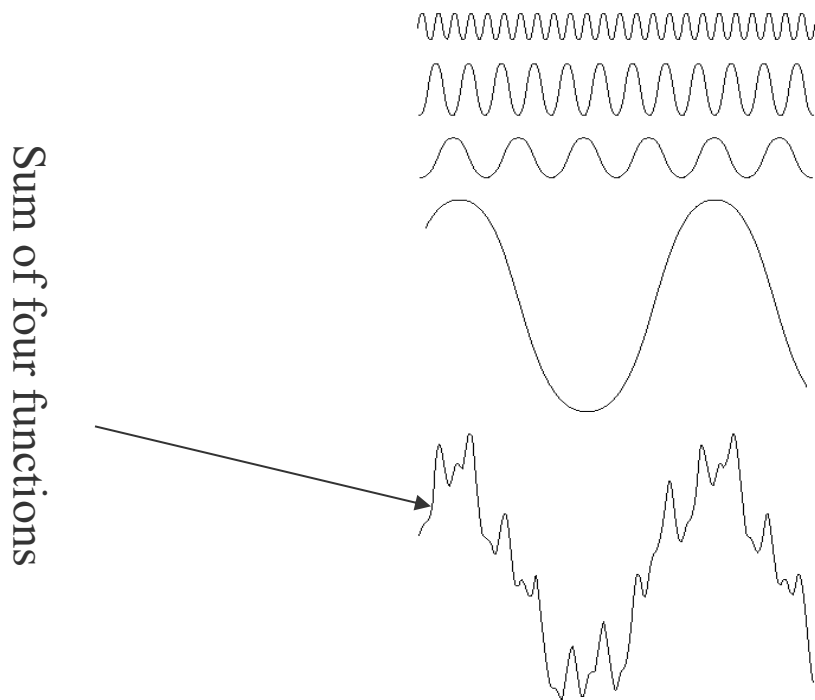


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

Fourier Transform (1 of 5)

- ❖ The Fourier transform of $F(u)$ of a single variable, continuous function, $f(x)$, is

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx$$

- ❖ The Fourier transform of $F(u,v)$ of a double variable, continuous function, $f(x,y)$, is

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy$$

Fourier Transform (2 of 5)

- ❖ The Fourier Transform of a discrete function of one variable, $f(x)$, $x=0,1,2,\dots,M-1$, is,

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x) e^{-2j\pi ux/M}$$

where, $u=0,1,2,\dots,M-1$,

- ❖ The concept of Frequency domain follows Euler's formula:

$$e^{j\theta} = \cos \theta + j \sin \theta$$

Fourier Transform (3 of 5)

- ❖ Fourier Transform (in one dimension)

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x) [\cos 2\pi ux / M - j \sin 2\pi ux / M]$$

Where, for $u = 0, 1, 2, \dots, M$

- ❖ Each term of the Fourier transform is composed of the sum of all values of the function $f(x)$.
- ❖ The values of $f(x)$ are multiplied by sine and cosines of various frequencies
- ❖ Each of the M term of $F(u)$ is called the frequency component of the transform
- ❖ The domain (values of u) over which the values of $F(u)$ range is appropriately called the frequency domain

Fourier Transform (4 of 5)

- ❖ Fourier Transform (in two dimension)

$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

- ❖ It is common practice to multiply the input image function by $(-1)^{x+y}$ prior to computing Fourier transform. This shifted the origin of the transform to $u=M/2$ and $v=N/2$

Fourier Transform (5 of 5)

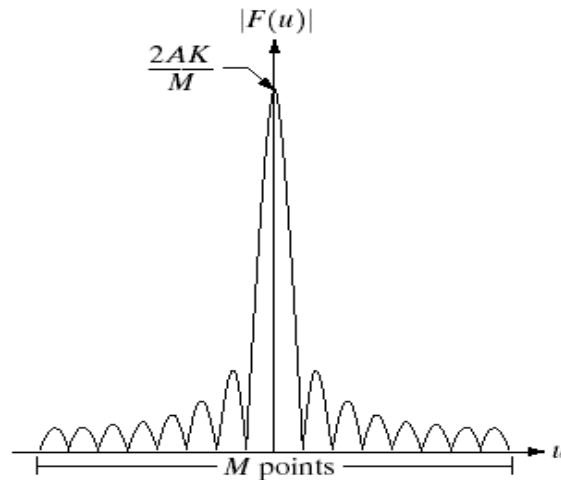
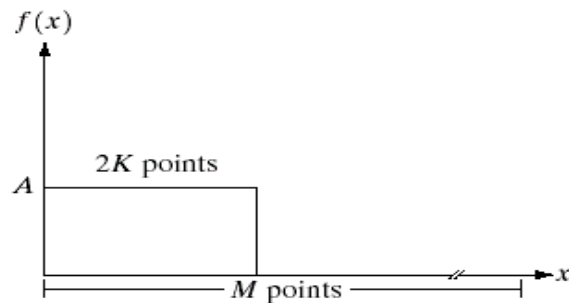
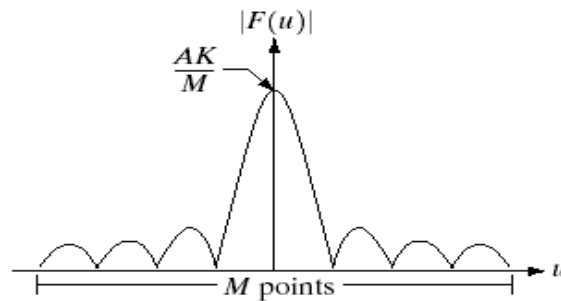
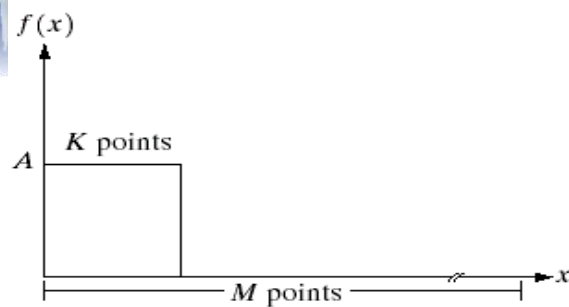
- ❖ The value of the transform at $(u,v) = (0,0)$ is

$$F(u, v) = \frac{i}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)$$

Which we see is the average of $f(x, y)$.

- ❖ In other words if $f(x,y)$ is an image the value of the Fourier transform at the origin is equal to the average gray level of the image.

Spatial Vs. Frequency Domain

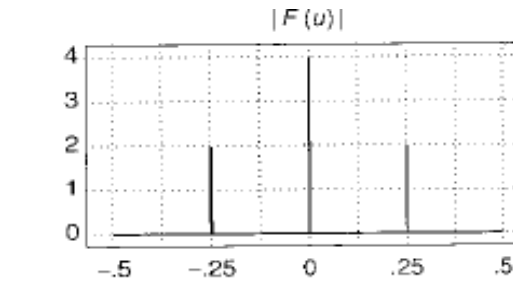
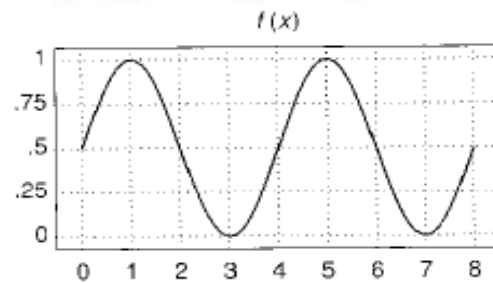


a	b
c	d

FIGURE 4.2 (a) A discrete function of M points, and (b) its Fourier spectrum. (c) A discrete function with twice the number of nonzero points, and (d) its Fourier spectrum.

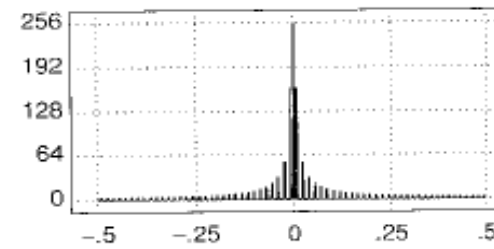
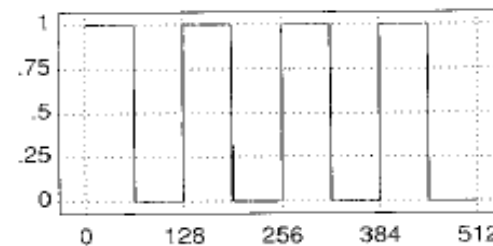
Spatial Vs Frequency Domain

Sine wave

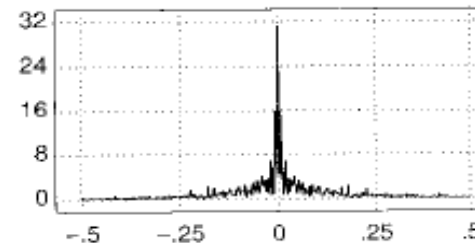
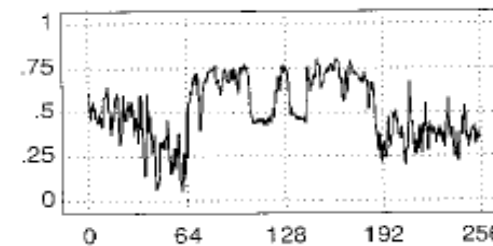


(a)

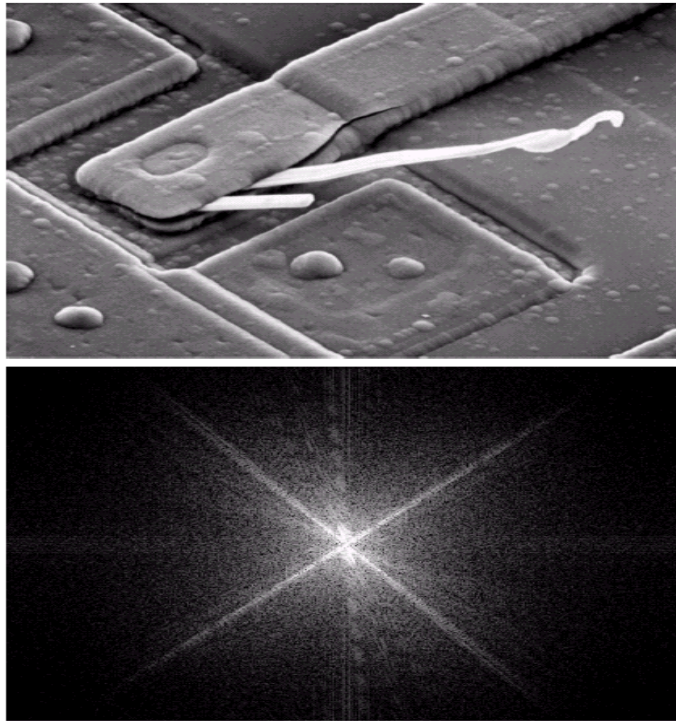
Square wave



(b)



An Image and its Fourier Spectrum



a
b

FIGURE 4.4

(a) SEM image of a damaged integrated circuit.
(b) Fourier spectrum of (a).
(Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

**SEM=Scanning Electron
Microscope**

- ❖ Rate of change in frequency in FT is equal to the pattern of intensity variation.
- ❖ Away from origin higher the frequency and faster gray level change.

Image Filtering in the Frequency Domain

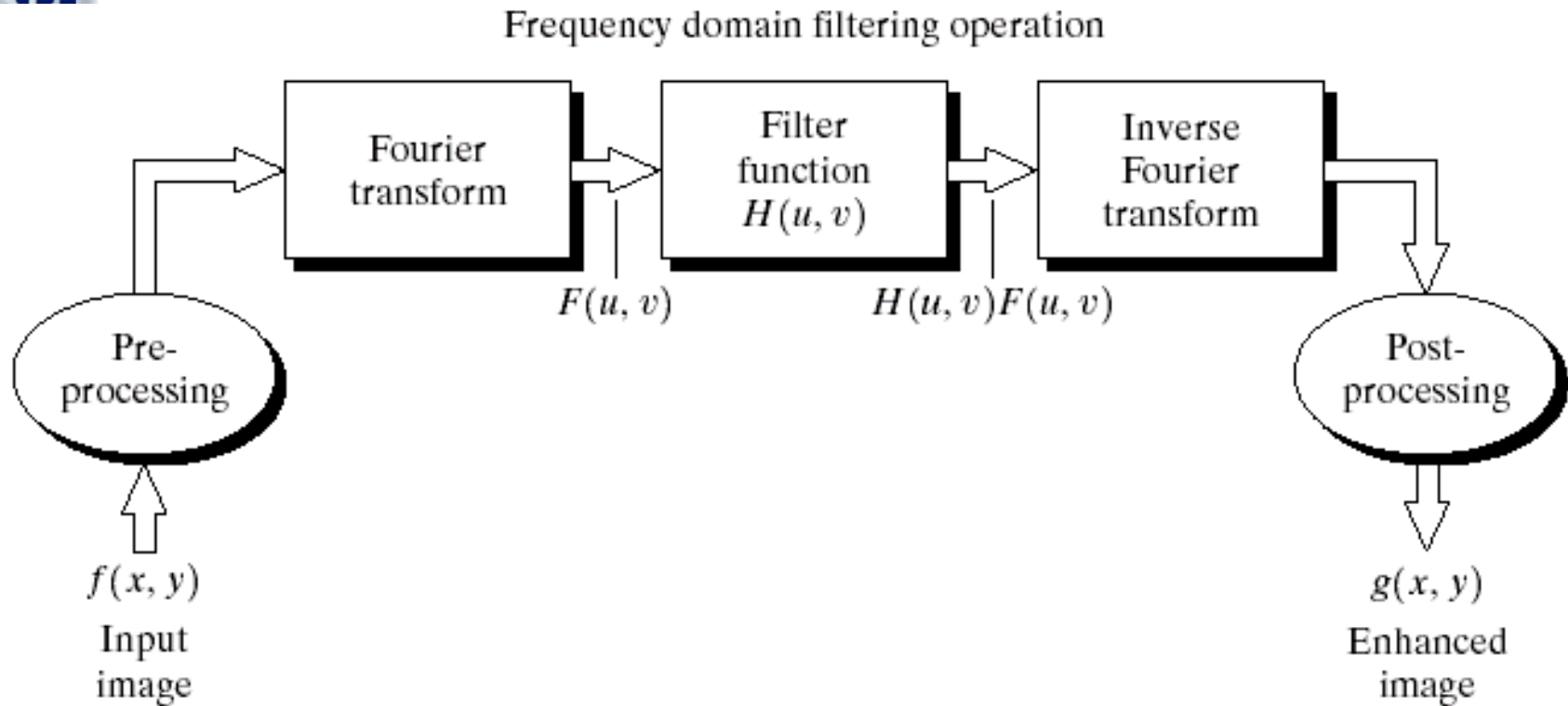


FIGURE 4.5 Basic steps for filtering in the frequency domain.

Filtering in the Frequency Domain

❖ Notch Filter (lowering the average gray level)

$$H(u, v) = \begin{cases} 0 & \text{if } (u, v) = (M / 2, N / 2) \\ 1 & \text{otherwise} \end{cases}$$

❖ $G(u, v) = H(u, v) F(u, v)$



Filtering in the Frequency Domain

- Low pass filtering

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

$$D(u, v) = \left[(u - M/2)^2 + (v - N/2)^2 \right]^{1/2}$$

- Cuts off all high frequency over D_0

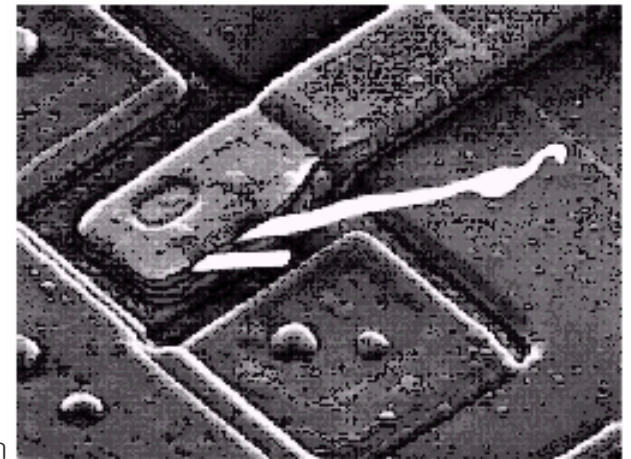


Filtering in the Frequency Domain

High Pass Filter

$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$

Cuts off all frequency below D_0





List of Reference Books

- ❖ Rafael C. Gonzalez, “Digital Image Processing”.
- ❖ Anil K. Jain, “Fundamental of Digital Image Processing”.
- ❖ Dana H. Ballard, “Computer Vision”
- ❖ David H. Hubel, “Eye, Brain, and Vision”
- ❖ Dwayne Phillips, “Image Processing in C”